Team Details

Team Name: bunker_backer

SR. NO	ROLE	NAME	ACADEMIC YEAR
1	**Team Leader**	*Y NAVADEEP SARAN*	*THIRD (III)*
2	Member 1	*G SHANMUKH SAI*	*THIRD (III)*
3	Member 2	*S SAHITH SOMASUNDAR*	*THIRD (III)*
4	Member 3	*NADIMPALLI VANISHA*	*THIRD (III)*

A team can have up to 4 members including the team leader. Add rows if necessary.

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Problem Statement Addressed

Selected the problem statement your idea addresses

DESCRIPTION / DETAILS

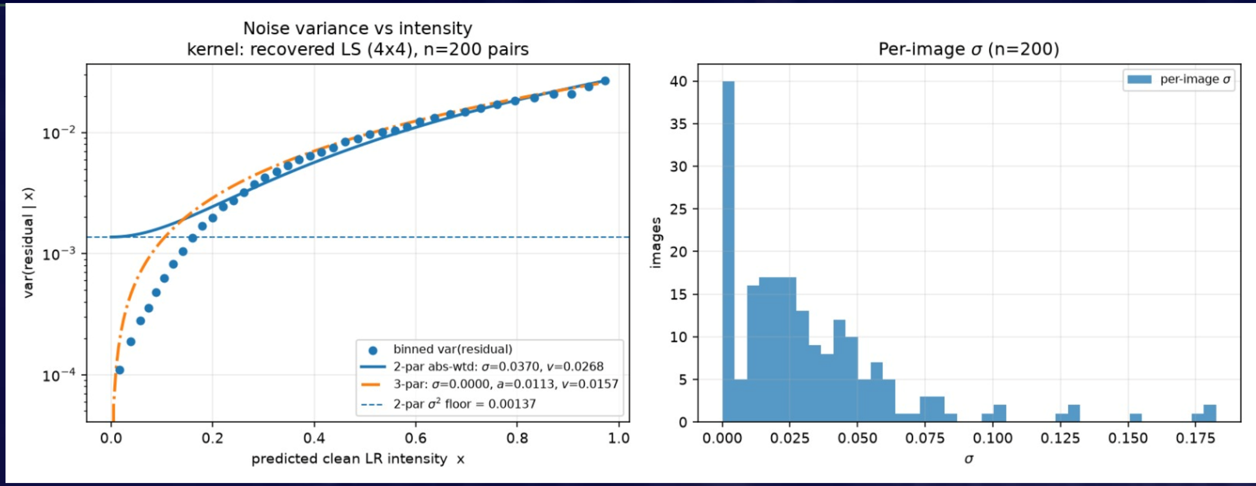
KLA PS01 — AI-Based Restoration of Degraded Images for Semiconductor Inspection.
A single lost pixel or noisy region in an inspection image can hide a real defect and cost a die — restoration quality is a correctness problem, not a cosmetic one.

The task: recover a clean, full-resolution image from a version that has been downsampled by exactly 2x AND corrupted by two kinds of signal-dependent noise(speckle and additive/shot noise), applied together in an order that is not disclosed.The model must do this blindly, in one pass, and generalise to content it never saw during training.

The released dataset is 3200 training pairs and 400 test inputs of grayscale natural photographs, not semiconductor imagery. We treat it as a proxy: the degradation — ×2 decimation plus signal-dependent noise — is what transfers to inspection imagery, so we characterised the degradation empirically and optimised for degradation robustness rather than fitting content-specific priors.

Idea Description – Describe your Idea/Solution/Prototype

Provide a brief summary of your idea, including the key concept and approach. Provide a brief overview of your solution and how it addresses the problem statement.



Noise_variance_vs_intensity.png — plot proving the noise model was measured from real data, not guessed.

KEY CONCEPT & APPROACH

Restore the image in a single blind forward pass at low resolution, using one neural network that jointly reverses both the noise and the downsampling —no multi-stage cascade, no separate denoise-then-upscale pipeline. The degradation itself (the exact downsampling kernel and the noise's statistical behaviour) was measured directly from the provided data rather than assumed from the problem brief, and that measurement drives how training data is synthesised.

SOLUTION OVERVIEW

A compact convolutional network (NAFSR — NAFNet-style blocks with channel attention, 1.39M parameters) takes the degraded low-resolution image and outputs a clean image at double the resolution via a single upsampling head. It is trained to invert the measured real degradation, augmented with procedurally generated structural content, so it also generalises to patterns unlike anything in the natural-photo training images.

Proposed Solution - Describe your Idea/Solution/Prototype

Describe your idea in detail. Include the methodology, technologies involved, and how it addresses the chosen problem statement.

SOLUTION DETAILS

Pipeline: load input -> group same-shaped images into batches -> run through the network in reduced precision (bf16) for speed -> clip output to a valid[0,1] range -> save.

Architecture: NAFSR, NAFNet-style blocks (SimpleGate activation, simplified channel attention, LayerNorm), plus a PixelShuffle upsampling head for the2x resolution increase. Includes noise-level conditioning (the network is told how noisy the input roughly is) and an uncertainty output (the model flags which regions it's less confident about).

Loss function: a balanced mix of pixel-level fidelity, structural similarity , and frequency-domain accuracy — deliberately no adversarial/GAN loss, because inventing plausible-looking but incorrect detail is the worst possible failure mode for an inspection use case.

Training data: real training pairs plus synthetic pairs generated on the fly using the measured degradation model, plus procedurally generated structural patterns (gratings, grids, checkerboards, circuit-trace-like shapes) mixed in to improve generalisation to structured content.

Innovation and Uniqueness

Highlight what makes your idea unique or innovative compared to existing solutions.

KEY INNOVATION

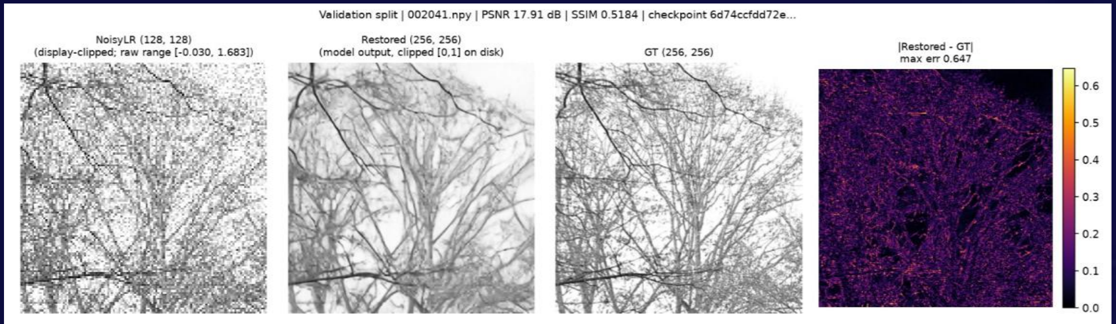
The downsampling kernel and noise behaviour were not assumed from the problem brief — they were measured empirically from the provided data(least-squares kernel recovery, noise-variance-vs-intensity curve fitting),and that measurement directly drives the synthetic training-data generator. This is different from typical approaches that assume a generic degradation model.

COMPETITIVE ADVANTAGE

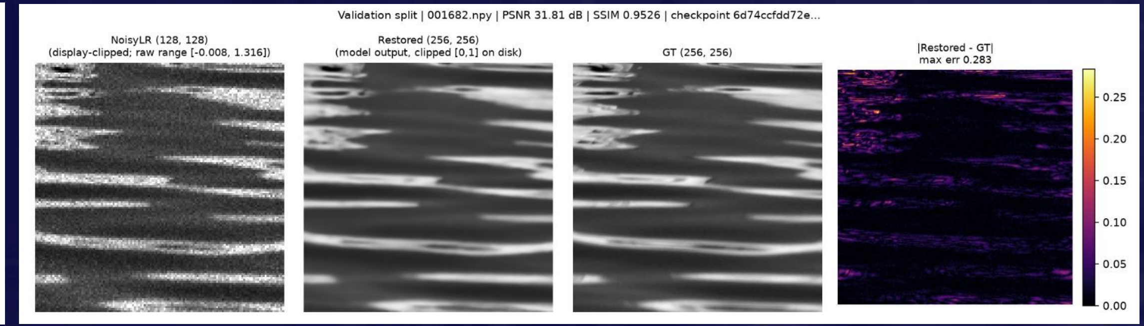
No hallucination risk: the model is explicitly trained and verified to blur rather than invent detail on content it cannot reliably recover — confirmed directly by comparing the frequency content of its output against what the input can actually support. Every claimed number in this project (accuracy , speed, generalisation) comes from a script that can be re-run to reproduce it, rather than being manually reported.

Impact and Benefits

Explain how your solution will make an impact, such as improving performance, reducing costs, increasing efficiency, or solving other challenges.



val_002041_weak_worst_in_split_psnr17.91.png — the honest worst-case failure, shown deliberately.



Eval_001682_typical_near_mean_psnr31.81.png — a typical, successful restoration (before/after).

Primary Impact

A more accurate and more honestly-validated restoration reduces the risk of a real defect being hidden by image degradation during inspection, and reduces false confidence from a model that would otherwise invent detail where none exists.

Quantifiable Outcomes

+5.9 dB PSNR over a simple bicubic upscale, and +0.7 dB / better perceptual quality (LPIPS) over a comparable baseline neural network, on a 400-imageheld-out test — all differences statistically significant, not noise.~19-23 images per second end-to-end on a laptop GPU, depending on method

Technology & Feasibility/Methodology Used

Describe the technologies, methodologies, or tools you plan to use to implement your idea.

IMPLEMENTATION STRATEGY

Built and trained in PyTorch. Training ran on a cloud NVIDIA A100 GPU(rented per-hour, not owned hardware); inference and all speed measurements were run on a local NVIDIA RTX 4060 laptop GPU. The final model is1.39 million parameters — small enough to run quickly, verified to run on GPU with no internet access, no API keys, and no manual setup steps beyond installing pinned dependencies.

Software Architecture: PyTorch, NumPy — a single standalone Python script handles all inference, no external services

Hardware Components: Training: cloud NVIDIA A100 GPU. Inference/testing: NVIDIA RTX 4060 Laptop GPU (8GB).

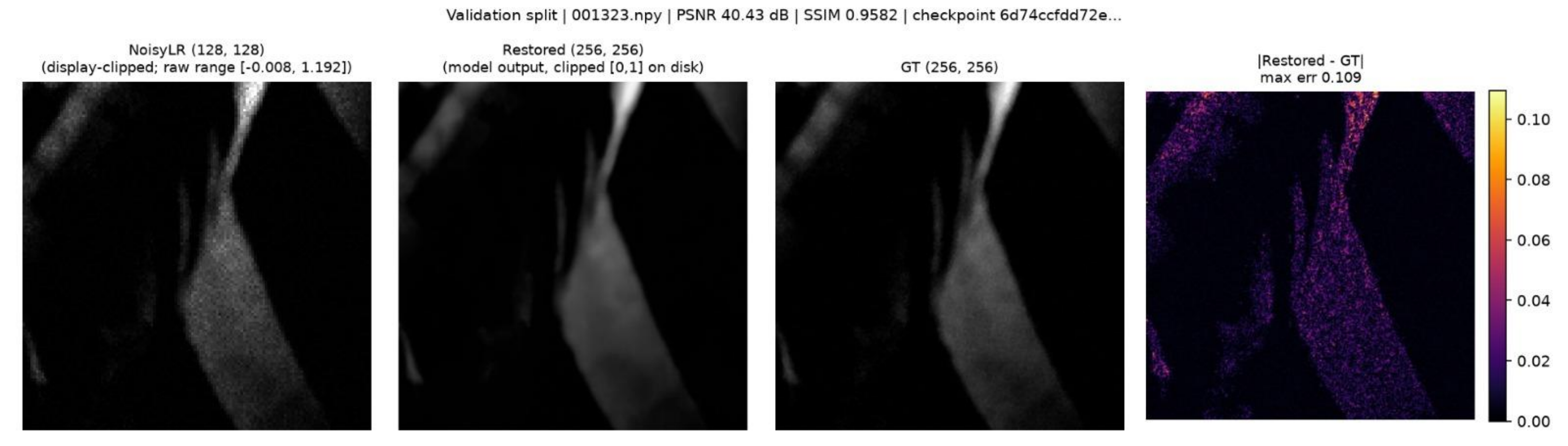
Development Tools: Python 3.12, CUDA 12.8, Git/GitHub for version control.

GitHub & Video Link

GitHub Repository

https://github.com/sahithsundarw/bunker_backer

Static example — best-scoring restoration (40.43 dB PSNR).



No demo video recorded — optional per submission guidelines. For proof of the pipeline working end-to-end, see sample_inputs/ and results/sample_outputs/ in the repository, or run run.py directly.

Research and References

Research Background & Methodology

Briefly describe the research foundation or scientific principles supporting your idea.

The approach builds on established image-restoration and super-resolution research: NAFNet-style efficient convolutional blocks for restoration ,PixelShuffle for learned upsampling, SSIM and LPIPS as perceptual quality measures, and data-augmentation techniques adapted for super-resolution tasks. The degradation model itself was derived empirically from the provided dataset rather than taken from any single paper.

References & Citations

List key papers, articles, or data sources.

Chen, Liangyu, et al. "Simple baselines for image restoration." European conference on computer vision. Cham: Springer Nature Switzerland, 2022.

Shi, Wenzhe, et al. "Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network." Proceedings of the IEEE conference on computer vision and pattern recognition. 2016.

Zhang, Richard, et al. "The unreasonable effectiveness of deep features as a perceptual metric." 2018 IEEE/CVF conference on computer vision and pattern recognition. IEEE, 2018.